Robust Dedicated Actuator Placement via Matroid Spanning

Qichao Wang[1] Daoyu Dong[2]

[1]Hebei University of Technology
[2]University of Electronic Science and Technology of China

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Abstract

We study dedicated actuators installed at distinct states of a structured linear system, where failures remove actuators but leave all states and state edges intact. For unit costs and one failure, an exact source-quota relaxation followed by maximum matching in a common-repair graph gives a 3/2-approximation on arbitrary directed state graphs in $O(n^3)$ graph operations. The proof applies the Tutte–Berge formula to disjoint unions of fundamental cocircuits. For multiple failures, we prove exact fixed-parameter tractability in the maximum-matching deficiency plus the failure budget, independently of the number of source strongly connected components. The underlying theorem optimizes robust matroid spanning with arbitrarily many disjoint unit-cost quotas. We also prove a weighted safe/vulnerable spanning theorem in the independence-oracle model. The latter extends to control on strongly connected graphs, but does not assert weighted optimization with arbitrary source quotas. Complete proofs and explicit algorithmic bounds are provided. Publication priority and the optimal polynomial-time approximation ratio remain unresolved.

1 Introduction

A structurally controllable system can lose controllability when an actuator fails. Redundant placement must protect both the matching condition and accessibility from the installed inputs. These requirements have different combinatorial forms: robust spanning in a matroid and lower quotas on source strongly connected components. Keeping both requirements explicit permits approximation and exact parameterized algorithms within the same model.

Our first result is a cubic-time 3/2-approximation for one actuator failure and unit costs (Theorem 3.3). Its repair argument is valid for arbitrary matroids. The exact graph relaxation, not a minimum ordinary controllability placement, provides the necessary lower bound. A three-state example attains the ratio for a permissible relaxation choice; this is not an inapproximability result.

Our second result concerns arbitrary independence-oracle matroids with nonnegative purchase weights and safe or vulnerable elements (Theorem 4.1). Building on the uniform-set core method of Bentert et al. [5], a weight-ordered recursion preserves an optimum robust purchased set. The recursion continues into lower-rank closures even when a large uniform set has been found. A classical set-pairs inequality sharpens the deficient-hyperplane count. The parameter is rank plus failure budget, and the objective charges every purchased element.

The third result is exact unit-cost optimization with arbitrarily many disjoint quota groups (Theorem 5.1). A bounded robust core need not contain all elements needed for accessibility. We

instead preserve its quota-completion cost by capping the number of representatives of each group. The resulting control algorithm is FPT in matching deficiency plus failure budget on arbitrary digraphs (Corollary 5.2); the number of source components is not a parameter. This quota theorem does not assert a weighted extension.

Poças, Pequito and Ramos [13] study the robust minimal structural controllability problem, establishing NP-hardness and reporting approximation factors 5/2 for general digraphs and 2 for strongly connected digraphs. We state every model convention below rather than identifying distinct input models solely by their problem names. Related robust-assignment algorithms and fault-tolerant matroid results are discussed in Section 6. No first-result or best-known claim is made.

2 Model and the dual matching matroid

Let $G = (E, A)$ be a nonempty finite directed state graph with $n = |E|$; self-loops are allowed. Each prescribed nonzero entry of the state and input matrices is a separate free parameter. Structural controllability means that a controllable numerical realization exists, equivalently that almost every realization is controllable. It is not strong structural controllability. A placement $P \subseteq E$ installs one dedicated input at each selected state, at most one per state. Failure removes that input only. It does not remove its state vertex or any state edge. There are no unpurchased standby actuators available after a failure.

Write $S_1, \dots, S_s$ for the source strongly connected components (SCCs), those having no entering state edge from another SCC. Form the state bipartite graph with a left and right copy of E, and an edge $u_L v_R$ for each arc $u \to v$. Let T be the transversal matroid on the right copies: its independent sets are precisely the state sets matchable using state edges. Put $M = T^*$ and $d = n - \operatorname{rank}_T(E)$, the maximum-matching deficiency. For $X \subseteq E$, duality gives

$$\operatorname{rank}_M(X) = |X| + \operatorname{rank}_T(E \setminus X) - \operatorname{rank}_T(E). \tag{1}$$

In particular, X spans M if and only if $E \setminus X$ is independent in T. A rank oracle for M is computable by bipartite matching.

The standard structural-controllability criterion combines accessibility with a matching saturating all state vertices [9, 12]. With dedicated inputs it yields the following useful formulation.

Proposition 2.1. *A placement P tolerates every set of at most k actuator failures if and only if*

$$\operatorname{rank}_M(P \setminus F) = d \quad (F \subseteq P,\ |F| \le k), \qquad |P \cap S_i| \ge k + 1 \quad (1 \le i \le s). \tag{2}$$

Proof. For a fixed surviving placement X, matching every state in $E \setminus X$ by state edges and every state in X by its own input gives a matching saturating all states. Conversely, an input-augmented saturating matching matches every state in $E \setminus X$ using state edges. This proves the matching equivalence, and (1) gives spanning in M. Accessibility holds exactly when every source SCC contains an input: every component is reachable from a source component in the condensation DAG, and no input outside a source component can reach it. The source SCCs retain an input after every at most k failures exactly when they initially contain at least $k + 1$. Applying both criteria to every survivor proves the claim. □

We use standard finite-matroid notation. The closure of X is $\operatorname{cl}(X) = \{e : \operatorname{rank}(X \cup \{e\}) = \operatorname{rank}(X)\}$. A hyperplane is a flat of rank one below the matroid rank. A loop has rank zero; a coloop belongs to every basis. The coloops of a restriction $M|P$ are precisely the elements whose deletion lowers $\operatorname{rank}_M(P)$. A circuit is a minimal dependent set. For a basis B and $e \notin B$, the unique circuit contained in $B \cup \{e\}$ is its fundamental circuit, including the one-element circuit when e is a loop. We use these elementary matroid properties [11].

3 A cubic time three halves approximation

Throughout this section $k = 1$ and every actuator has unit cost.

3.1 An exact source quota relaxation

Minimize $|P|$ subject to P spanning M and $|P \cap S_i| \geq 2$ for all i. Denote its optimum by L. Every robust placement is feasible for this relaxation, so $L \leq \mathrm{OPT}$.

It is solvable by an integral maximum flow. The complement $I = E \setminus P$ must be matchable using state edges and must satisfy $|I \cap S_i| \leq |S_i| - 2$. Use a source, the two state copies, one group node per source SCC, and a sink. Source-to-left-state edges have capacity one; left-to-right state edges have capacity one. Each right state has a capacity-one outgoing edge, to its source group if it has one, and directly to the sink otherwise. The edge from group i to the sink has capacity $|S_i| - 2$. An integral maximum flow gives a maximum-cardinality admissible I, and therefore a minimum-cardinality P.

A singleton source SCC makes robust placement impossible. Conversely, if all source SCCs have at least two vertices, every vertex has an incoming state edge: this is immediate within a nonsingleton SCC, and a singleton nonsource SCC has an entering edge. Thus T has no loops and M has no coloops. The full placement E spans after every single deletion and satisfies all quotas. This proves that singleton source SCCs characterize infeasibility for this one-failure model.

3.2 Cocircuit unions and a global lower bound

The next two lemmas concern an arbitrary rank-d matroid M with a spanning set P. Let D be the coloops of $M|P$. For $b \in D$ define

$$C_b = E \setminus \mathrm{cl}(P \setminus \{b\}).$$

The set C_b consists of the elements that restore spanning after deleting b. It meets P exactly in $\{b\}$.

Lemma 3.1. *For every $R \subseteq D$,*

$$\mathrm{rank}\left(E \setminus \bigcup_{b \in R} C_b\right) = d - |R|.$$

Proof. Choose a basis $B \subseteq P$. Every $b \in D$ belongs to B, and $B \setminus \{b\} \subseteq P \setminus \{b\}$ are both of rank $d - 1$. Their closures are consequently equal. For nonempty R we claim

$$\bigcap_{b \in R} \mathrm{cl}(B \setminus \{b\}) = \mathrm{cl}(B \setminus R).$$

For $e \notin B$, membership in $\mathrm{cl}(B \setminus \{b\})$ means that its fundamental circuit omits b. Membership for every $b \in R$ therefore means that this circuit is contained in $(B \setminus R) \cup \{e\}$, which is precisely membership in $\mathrm{cl}(B \setminus R)$. Basis elements satisfy the equality directly. This also handles loops. Taking ranks proves the identity. For $R = \varnothing$ it is simply $\mathrm{rank}(E) = d$. □

Form a simple undirected graph K on D. Put an edge bc when some $e \in E \setminus P$ belongs to both C_b and C_c, and retain one such witness for each edge. This graph is not necessarily bipartite. Let ν be its maximum matching cardinality.

Lemma 3.2. *Every set W spanning M after any one deletion satisfies $|W| \geq 2(|D| - \nu)$.*

Proof. Fix $U \subseteq D$ and let $R_1, \ldots, R_t$ be the nonempty connected components of $K - U$. Put $Q_i = \bigcup_{b \in R_i} C_b$. These sets are pairwise disjoint. An element of P belongs to at most one C_b. A common external element of Q_i, Q_j would witness an edge between their components. Lemma 3.1 gives $\operatorname{rank}(E \setminus Q_i) = d - |R_i|$. Since adding an element raises rank by at most one, W must contain at least $|R_i|$ elements of Q_i to span. Equality is impossible: deleting one of these elements would leave rank at most $(d - |R_i|) + (|R_i| - 1) < d$. Thus

$$|W| \geq \sum_i (|R_i| + 1) = |D| - |U| + t \geq |D| - |U| + \operatorname{odd}(K - U),$$

where odd counts odd-order components. The classical Tutte–Berge formula [10] states

$$2\nu = |D| - \max_{U \subseteq D} \bigl(\operatorname{odd}(K - U) - |U|\bigr).$$

Taking the maximum in the preceding inequality proves the result. The argument compares against every robust W, not just supersets of P. □

3.3 Algorithm and guarantee

Theorem 3.3. *For any nonempty directed state graph in the stated dedicated-actuator model, a robust placement of size at most $3\,\mathrm{OPT}/2$ can be found, or infeasibility reported, in $O(n^3)$ graph operations and $O(n^2)$ space. The guarantee is for unit costs and one actuator failure.*

Proof. Reject singleton source SCCs; otherwise compute an optimum relaxation P. Find D and a maximum matching of K. For each matched pair install its external common-repair witness. For every unmatched $b \in D$, install any external element of C_b. Such an element exists because M has no coloops, while $P \setminus \{b\}$ has rank $d - 1$. Let R be the set of installed witnesses, with repetitions removed. There are at most $\nu + (|D| - 2\nu) = |D| - \nu$ witnesses.

The set $P \cup R$ is robust. If a witness fails, P still spans. If an original noncritical element fails, the surviving part of P spans. If $b \in D$ fails, its witness repairs the deficiency. The source quotas persist under additions. By Lemma 3.2,

$$|P \cup R| \leq L + |D| - \nu \leq \mathrm{OPT} + \frac{\mathrm{OPT}}{2}.$$

Witness reuse is legitimate and can only decrease the cardinality. When $D = \varnothing$, P itself is robust.

For the complexity, retain the state matching J obtained from the quota flow. It covers $I = E \setminus P$. For each $b \in P$, explore alternating paths from unmatched right vertex b: traverse state edges to the left, then matched edges back to the right. A reachable free left vertex is exactly an augmenting path showing that $I \cup \{b\}$ is matchable, or that b is noncritical. For a critical b, the reachable matched right vertices are exactly its external repairs. A path to right vertex e can be flipped to match $I \setminus \{e\} \cup \{b\}$. Conversely, the symmetric difference with any matching of that replacement set gives such a path, since a path to a free left vertex would contradict criticality. Hence these searches recover the complete repair supports.

There are $O(n)$ vertices and $O(n^2)$ edges in the flow network. At most n unit augmentations, each found in $O(n^2)$ operations, suffice. At most n alternating searches also cost $O(n^3)$. For each external state, connect all pairs of critical states it repairs, storing a witness for each edge. This clique construction costs $O(n^3)$. A maximum general-graph matching is computable in $O(n^3)$ operations by a blossom algorithm [10]; a maximal matching is insufficient. All structures require $O(n^2)$ space. These are combinatorial operation bounds; the reference Python bitmask implementation has the usual additional integer bit costs. □

For the bidirected three-state star with center a and leaves b, c, the relaxation can choose $P = \{a, c\}$. Its one critical state is c, and the only external repair is b. The algorithm returns all three states, while $\{b, c\}$ is robust. Thus the ratio 3/2 is attained for this relaxation choice. Choosing the optimum robust pair already at the relaxation stage avoids the example. It does not rule out improved algorithms or establish 3/2-hardness.

4 Weighted robust matroid spanning

We now leave the unit-cost control model and state an abstract theorem. Let $M = (E, \mathcal{I})$ be a rank-r matroid given by an independence oracle, let $V \subseteq E$ be its vulnerable elements, and let $k \geq 0$. A set B is *resilient* if

$$\operatorname{rank}_M(B \setminus F) = r \quad \text{for every } F \subseteq B \cap V, \quad |F| \leq k. \tag{3}$$

Elements in $E \setminus V$ are safe. With nonnegative binary-encoded weights w, minimize $w(B) = \sum_{e \in B} w(e)$. Failures concern the selected set, not survival of the whole ambient matroid.

Theorem 4.1. *An optimum resilient purchased set can be computed, or infeasibility reported, in $f(r,k)\,\mathrm{poly}(n, L)$ time and oracle calls, where L is the input bit length and independence queries have unit cost. For $r \geq 2$, writing $q = (k+1)r$, a bound is*

$$2^{O(qr^2(r+k))}\,\mathrm{poly}(n, L),$$

hence $2^{O((r+k)^5)}\,\mathrm{poly}(n, L)$. The cases $r \leq 1$ or $k = 0$ are polynomial-time solvable.

The retained ground set has parameter-bounded cardinality and keeps access to the original oracle; this is not a polynomial-size encoded-kernel claim.

4.1 Matroid preliminaries and a small optimum

Loops can be discarded. Removing them from a resilient set preserves resilience, and their nonnegative weights cannot make this removal costly. If $r = 0$, the empty set is optimal. Henceforth assume $r \geq 1$ and work on the loop-free restriction. Set

$$q = (k+1)r, \qquad H = \binom{r+k-1}{k}, \qquad t_h = (h-1)H + q \quad (1 \leq h \leq r). \tag{4}$$

The same global values of q, H are used in every recursive subproblem.

Lemma 4.2. *Every resilient set B contains a resilient subset B_0 of cardinality at most q. Consequently, if a feasible solution exists, there is a minimum-weight solution of cardinality at most q.*

Proof. Successively choose a maximal independent subset I_j of $B \setminus \bigcup_{i<j} I_i$, for $j = 0, \ldots, k$, and put $B_0 = \bigcup_{j=0}^{k} I_j$. Empty layers are allowed. Since $|I_j| \leq r$, we have $|B_0| \leq q$.

Fix $F \subseteq B_0 \cap V$ with $|F| \leq k$. The layers are disjoint, so some I_j avoids F. Every element of $B \setminus B_0$ was available when I_j was chosen and belongs to $\mathrm{cl}(I_j)$. Every element of $B_0 \setminus F$ already lies in that surviving set. Therefore

$$B \setminus F \subseteq \mathrm{cl}(B_0 \setminus F).$$

Since F is also an allowed failure of B, its left-hand side spans M. Thus B_0 is resilient. Nonnegative weights give $w(B_0) \leq w(B)$, and applying this construction to an optimum proves the second assertion. □

4.2 A weight respecting exchange

We first sharpen the deficient-hyperplane count using the classical set-pairs inequality of Bollobás [6]. The short proof is included; no representation or bound on the size of Y is needed.

Lemma 4.3. *For any $Y \subseteq E$, at most $\binom{r+k-1}{k}$ distinct hyperplanes can have the form $\mathrm{cl}(Y \setminus F)$ with $F \subseteq Y$ and $|F| \le k$. This count is sharp.*

Proof. Index these hyperplanes by P_i. Choose a basis $A_i \subseteq Y \cap P_i$ of size $r-1$, and put $D_i = Y \setminus P_i$. Because $P_i = \mathrm{cl}(Y \setminus F)$ for some permitted F, we have $D_i \subseteq F$, so $|D_i| \le k$. Also $A_i \cap D_i = \varnothing$. If $i \neq j$, then $A_i \cap D_j \neq \varnothing$: otherwise $A_i \subseteq P_j$ and hence $P_i = \mathrm{cl}(A_i) \subseteq P_j$, forcing equality of the two hyperplanes.

In a uniformly random permutation of Y, let $\mathcal{E}_i$ be the event that every element of A_i precedes every element of D_i. Its probability is $\binom{r-1+|D_i|}{r-1}^{-1}$. These events are pairwise disjoint. Indeed, for $i \neq j$ choose $a \in A_i \cap D_j$ and $b \in A_j \cap D_i$. The two events require a before b and b before a, respectively. Therefore

$$\sum_i \binom{r-1+|D_i|}{r-1}^{-1} \le 1.$$

Each summand is at least $\binom{r+k-1}{r-1}^{-1}$, proving the bound. The argument includes $r = 1$ and $k = 0$, when there can be at most one such hyperplane.

For sharpness take $M = U_{r,r+k}$, $B = E$, all elements vulnerable, and $Y = B \setminus \{x\}$. The set B is resilient, and each k-subset of Y leaves a distinct $(r-1)$-element hyperplane. Thus equality holds even for the deficient hyperplanes needed below. This does not assert sharpness of the full core threshold t_h. □

Call Z *h-uniform* if $\mathrm{rank}(Z) = h$ and every h-element subset of Z is independent. Such a set has at least h elements, and every subset of size at most h is independent. A set has a single failure status if it lies entirely in V or entirely in $E \setminus V$.

Lemma 4.4. *Let $A \subseteq E$ have a single failure status and rank $h \ge 1$. Let $Z \subseteq A$ be h-uniform with $|Z| \ge t_h$. Suppose B is resilient, $|B| \le q$, and $x \in (B \cap A) \setminus Z$ satisfies $w(z) \le w(x)$ for every $z \in Z$. Then some $y \in Z \setminus B$ makes*

$$B' = (B \setminus \{x\}) \cup \{y\}$$

resilient, with $|B'| = |B|$ and $w(B') \le w(B)$.

Proof. Put $Y = B \setminus \{x\}$. Consider the permitted failures $F \subseteq Y \cap V$, $|F| \le k$, for which $\mathrm{rank}(Y \setminus F) < r$. Resilience of B implies that their surviving ranks equal $r-1$ and that $x \notin \mathrm{cl}(Y \setminus F)$. Lemma 4.3 bounds the number of these distinct hyperplanes by H. It applies also when x is safe and some deficient failure sets have size less than k.

Each of these hyperplanes contains at most $h-1$ elements of Z. Otherwise it contains an independent h-subset of Z, which spans A, and hence contains x, a contradiction. At most $q-1$ elements of Z belong to B, because $x \notin Z$. The number of forbidden choices in Z is at most $(h-1)H + q - 1 < t_h$. Choose a $y \in Z \setminus B$ outside all the deficient hyperplanes.

Let $F' \subseteq B' \cap V$ have size at most k. If $y \notin F'$, then $F' \subseteq Y$; either $Y \setminus F'$ already spans or y repairs its rank deficiency by construction. If $y \in F'$, then y is vulnerable, so x is vulnerable as well. The set

$$(F' \setminus \{y\}) \cup \{x\}$$

is an allowed failure of B of the same size as F'. Its survivor is exactly $B' \setminus F'$, which therefore spans. The cardinality and weight assertions follow from the choice of y. □

The argument includes $r = 1$, when a deficient hyperplane has rank zero and its basis is empty. It does not assume distinct weights, positive weights, or representability.

4.3 Recursive construction of an optimum preserving core

For a loop-free set A of a single failure status, define $\mathrm{Core}(A)$ by the following procedure. Every scan uses a fixed total ordering that refines nondecreasing weight.

1. If $A = \varnothing$, return $\varnothing$. Let $h = \mathrm{rank}(A)$. If $h = 1$, return the q cheapest elements of A, or all of A if it has fewer. If $|A| \le t_h$, return A.

2. Scan the elements of A in order, initially with $Z = \varnothing$. Accept an element x precisely when every subset of $Z \cup \{x\}$ of size at most h is independent. Stop when $|Z| = t_h$ or all elements have been scanned.

3. For every $T \in \binom{Z}{h-1}$, put $A_T = A \cap \mathrm{cl}(T)$ and recursively compute $\mathrm{Core}(A_T)$. Return

$$Z \cup \bigcup_{T \in \binom{Z}{h-1}} \mathrm{Core}(A_T).$$

In particular, step 3 is executed under *both* stopping conditions. Recursive domains need not be flats of the original matroid.

Until Z has size h, the acceptance rule is ordinary greedy independence. The final Z therefore has at least h elements and is h-uniform. Each child has rank exactly $h - 1$: it contains T and is contained in $\mathrm{cl}(T)$.

Lemma 4.5. *For every resilient B with $|B| \le q$ and every single-status loop-free set A, there is a resilient B' such that*

$$|B'| = |B|, \quad w(B') \le w(B), \quad B' \setminus A = B \setminus A, \quad B' \cap A \subseteq \mathrm{Core}(A).$$

Proof. Induct on $h = \mathrm{rank}(A)$. Empty domains and a return of all of A require no change. If $h = 1$ and $|A| > q$, an element x of $B \cap A$ outside the q cheapest elements can be replaced by an unused retained element y of no larger weight. Nonloop elements of a rank-one domain are parallel. When y survives, substitution preserves rank; when y fails, it and x are both vulnerable, and the failure mapping in Lemma 4.4 applies. Repeating gives the claim. Equivalently, use Lemma 4.4 with $h = 1$ and $|Z| = q$.

Suppose $h > 1$ and the scan is executed. Let $U = \bigcup_T A_T$. Every rejected scanned element x belongs to U. Indeed, at rejection a dependent subset of $Z \cup \{x\}$ contains x and at most $h - 1$ retained elements. Those retained elements are independent, so they span x. Extend them to an $(h-1)$-subset of the final Z, obtaining a child containing x. Also $Z \subseteq U$, since $h > 1$ and every element of Z belongs to an $(h-1)$-subset of Z.

If the scan is exhausted, $A = U$. Otherwise each element of $A \setminus U$ is unscanned and is at least as expensive as every element of Z. Apply Lemma 4.4 successively to the elements of $B \cap (A \setminus U)$. Each replacement lies in $Z \subseteq U$ and removes one selected element outside U. The resulting resilient set has its intersection with A inside U, unchanged cardinality and outside-A part, and no larger weight.

Fix the final output $W_A = \mathrm{Core}(A)$ and process the children $A_1, \ldots, A_m$ in any order. Apply the induction hypothesis to each child and the current solution. For the current solution define

$D(B) = (B \cap A) \setminus W_A$. Processing A_i changes only elements in A_i and leaves its selected elements in $\mathrm{Core}(A_i) \subseteq W_A$. Consequently,

$$D(B_{\mathrm{new}}) \subseteq D(B_{\mathrm{old}}) \setminus A_i.$$

Initially $D(B) \subseteq U$. After all children have been processed it is empty, regardless of overlaps between children. All operations preserve cardinality, hence the bound q, and preserve resilience without increasing weight. This establishes the induction. □

Apply the construction separately to the vulnerable and safe nonloop elements, and let W be the union of the two outputs. Their domains are disjoint. Lemmas 4.2 and 4.5 imply that W contains an optimum of cardinality at most q whenever the instance is feasible.

4.4 Size and running time

Put $K_0 = 0$, $K_1 = q$, and, for $2 \le h \le r$, define

$$K_h = t_h + \binom{t_h}{h-1} K_{h-1}.$$

By induction, a rank-h call returns at most K_h elements. The sequence is nondecreasing; in particular $|W| \le 2K_r$, or $|W| \le K_r$ if all elements are vulnerable.

For a convenient explicit bound put $T = t_r$ and $N = 2rT^{(r-1)^2}$. Each of the two recursion trees has at most r levels and branching at most T^{r-1}. Thus there are at most N nodes and $|W| \le NT$. At a node, before h elements are accepted, one independence query tests a candidate. Subsequently it suffices to test the h-subsets containing the candidate. This uses at most $O(n(1 + T^{r-1}))$ independence queries per node. Testing $x \in \mathrm{cl}(T')$ for a child-defining independent $(h-1)$-set T' uses one independence query to $T' \cup \{x\}$, with the membership case $x \in T'$ handled directly. Computing a domain's rank requires at most n queries. Sorting and set bookkeeping have polynomial input cost multiplied by a parameter-only factor.

Finally enumerate all $B \subseteq W$ with $|B| \le q$, and for each enumerate all allowed failure sets and compute its surviving ranks greedily. There are at most $(NT+1)^q$ candidates and at most 2^q failures per candidate. Choose a feasible candidate of minimum total weight. If there is none, optimum preservation proves infeasibility. Exact weight sums and comparisons have polynomial bit cost in the original weight encoding.

For $r \ge 2$, put $p = r + k$. Since $H \le 2^{p-1}$ and $q \le (p+1)^2$, we have $\log_2 T = O(p)$. Hence $NT = 2^{O(r^2 p)}$, and enumerating the candidates and failures gives $2^{O(qr^2p)}\,\mathrm{poly}(n, L)$. As $q = O(p^2)$, this is at most $2^{O(p^5)}\,\mathrm{poly}(n, L)$. For fixed $k \ge 1$, the sharper estimate $H = O_k(r^k)$ gives $T = O_k(r^{k+1})$, $NT = r^{O_k(r^2)}$, and running time $r^{O_k(r^3)}\,\mathrm{poly}(n, L)$. There is no enumeration with exponent r on n, and no pseudopolynomial dependence on weight values.

If $k = 0$, ordinary minimum-weight-basis greedy is optimal. If $r = 1$, an optimum consists of either one cheapest safe nonloop element or $k+1$ cheapest vulnerable nonloop elements, choosing the cheaper available option. These observations also handle large k without a special feasibility assumption. Nonnegative rational weights can be handled by exact comparisons and sums with polynomial bit complexity; the accompanying implementation accepts integers and exposes a rank-oracle interface, which is simulated from an independence oracle by the greedy procedure described above.

4.5 Why simpler weighted reductions do not suffice

Stopping at a large uniform set. Consider a rank-two truncation of a partition matroid with two parallel classes of size two having weights $(0, 0)$ and $(1, 1)$, and five further singleton classes each of weight 100. Let $k = 1$. Then $q = 4$, $H = 2$, and $t_2 = 6$. The weight-ordered uniform scan reaches the threshold while retaining only one element from each cheap parallel class. Its robust optimum is 101, whereas the full instance has optimum 2, attained by the two cheap pairs. Thus sorting the scan is insufficient if a successful threshold causes the recursion to stop. Our construction retains the cheap rejected elements through the lower-rank children.

Preserving cheapest ordinary bases after failures. Dey and Kavitha [7] study a preserver $S \subseteq E$ such that, for every $F \subseteq E$ with $|F| \leq k$, $M|(S \setminus F)$ contains a minimum-cost ordinary basis of $M|(E \setminus F)$. That property need not preserve the optimum total cost of a resilient purchased set.

For a five-element example, take a rank-two truncation with parallel classes A, B of size two and a singleton class C. Give each element of $A \cup B$ weight 2 and the element of C weight 3. For every zero or one ground-element failure, $A \cup B$ still contains an ordinary basis of cost 4, which is minimum in the full surviving matroid. Yet the robust optimum inside $A \cup B$ is 8, and the full instance has optimum 7, using one element from each class. Two successive cheapest residual bases also produce $A \cup B$. Thus this preserver construction does not replace the optimum-preserving recursion above.

4.6 An application to dedicated input placement

Consider a nonempty strongly connected state digraph with nonnegative actuator costs and specified safe or vulnerable locations. Keep the single-actuator-per-state and actuator-only failure conventions of Section 2. Its dual state matching matroid has rank equal to the deficiency d; Proposition 2.1 supplies the matching interpretation, also when only vulnerable actuators may fail.

Corollary 4.6. *Minimum-cost robust dedicated-input placement on a strongly connected state digraph, under the actuator-failure model just specified, is FPT in $d + k$, with nonnegative binary-encoded costs and safe/vulnerable actuator locations.*

Proof. If $d > 0$, every surviving spanning set in M is nonempty. Strong connectivity then ensures accessibility, so Theorem 4.1 applies. If $d = 0$, matching is automatic and accessibility requires a nonempty survivor. Choose the cheaper of one cheapest safe actuator and $k + 1$ cheapest vulnerable actuators, omitting unavailable options. □

For general state digraphs, additional source SCC constraints are necessary. They are not preserved by the weighted same-status exchange alone. Section 5 supplies a separate unit-cost quota theorem, not a general weighted extension.

5 Exact parameterized optimization with source quotas

The spanning requirement alone does not enforce accessibility in a general state digraph. We now retain the source-SCC quotas without adding their number to the parameter. Throughout this section every selected element is vulnerable and costs one.

Theorem 5.1. *Let M be a rank-d matroid on E, given by an independence oracle, and let $S_1, \dots, S_s$ be pairwise disjoint subsets of E. For $k \geq 0$, a minimum-cardinality set D satisfying*

$$\operatorname{rank}_M(D \setminus F) = d \quad (F \subseteq D,\ |F| \leq k), \qquad |D \cap S_i| \geq k+1 \quad (1 \leq i \leq s)$$

can be found, or infeasibility certified, in $f(d,k)$ times a polynomial in the input size and oracle cost. The parameter does not include s.

Set $b = k+1$, $G_0 = E \setminus \bigcup_i S_i$, and $q = bd$. An undersized group makes the instance infeasible. If $d = 0$, choosing b elements from each group is optimal. Assume henceforth that $d \geq 1$.

By Lemma 4.2, every robust spanning set contains a robust spanning subset of size at most q: take $k+1$ successive maximal independent sets from its remaining elements. Every omitted element is spanned by every layer, and one layer avoids any k failures. The union of the layers therefore remains spanning. This small core contains no loops, although loops remain available for quota completion. For a core C define

$$\phi(C) = |C \cap G_0| + \sum_{i=1}^{s} \max\{0, |C \cap S_i| - b\}.$$

Its cheapest quota completion has cardinality $bs + \phi(C)$ and remains robust. Conversely, a small core inside an optimum placement has completion cost no greater than that optimum. It suffices, therefore, to minimize ϕ over robust spanning sets of size at most q.

The ordinary robust core alone need not preserve completion cost. In $U_{1,4}$ on $\{a,b,c,d\}$, with $k = 1$ and the sole group $S_1 = \{c,d\}$, the robust core $\{a,b\}$ completes to four elements, whereas the robust optimum $\{c,d\}$ has size two. The following color argument retains the necessary group choices.

Exchange candidates. For this section use the conservative count $H = \binom{q-1}{d-1}$, matching the quota implementation. Lemma 4.3 also permits the smaller count $\binom{d+k-1}{k}$, but that refinement is not needed for FPT. Let A be a loopless rank-h subset and let $Z \subseteq A$ be h-uniform: all of its subsets of size at most h are independent. Assume $|Z| \geq h$. For a robust set B of size at most q and $x \in (B \cap A) \setminus Z$, write $Y = B \setminus \{x\}$. Whenever $\operatorname{rank}_M(Y \setminus F) < d$ for $F \subseteq Y$, $|F| \leq k$, the closure of $Y \setminus F$ is a hyperplane excluding x. There are at most H such hyperplanes, since each has a $(d-1)$-element basis in Y. Each contains at most $h-1$ points of Z; otherwise it would contain A and hence x. Thus at most $(h-1)H + q - 1$ points of Z lie either in a deficient hyperplane or in Y. Every other y gives a robust exchange $Y \cup \{y\}$: failures avoiding y are repaired by y, while a failure containing y corresponds to the failure of x in B.

A color-capped core. Start the recursion on the nonloop set $E^\circ = \{e \in E : \operatorname{rank}_M(\{e\}) = 1\}$. Keep all original elements, including loops, available for quota completion. Use $G_0, S_1, \dots, S_s$ as colors and set

$$c_h = (h-1)H + q, \qquad T_h = (q+1)c_h.$$

In a rank-h set A, greedily scan its elements into an h-uniform set Z, allowing at most c_h points of each color. Stop at $|Z| = T_h$ or at exhaustion. For $h = 1$ return Z. Otherwise return Z together with recursively computed cores of

$$A \cap \mathrm{cl}_M(T), \qquad T \in \binom{Z}{h-1}.$$

The recursion is performed in both stopping cases. The scan attains at least h points: before that the color cap cannot bind because $c_h \geq q \geq d \geq h$, so the scan is ordinary greedy independence. Every child consequently has rank exactly $h - 1$.

We prove the following strong invariant by induction on h: every robust B of size at most q can be replaced, without changing its size or its part outside A, by a robust B' whose A-part lies in the returned core and for which $\phi(B') \leq \phi(B)$. Let $U = Z$ for $h = 1$; otherwise let U be the union of the child ground sets $A \cap \mathrm{cl}_M(T)$. Then $Z \subseteq U$. A scanned point rejected for dependence lies in U: its witnessing dependent set shows that it is spanned by at most $h - 1$ retained points, which can be padded to one of the sets T.

Consider $x \in (B \cap A) \setminus U$. If it was scanned, it was rejected by its color cap. The c_h same-color points of Z contain a valid exchange candidate, since at most $c_h - 1$ are forbidden by the preceding count. The exchange preserves all color counts, and hence ϕ. If x was unscanned, the total threshold was reached. With $Y = B \setminus \{x\}$, forbid G_0 and all groups satisfying $|Y \cap S_i| \geq b$. There are at most q forbidden colors, since $|Y| \leq q - 1$ and $b \geq 1$. They contain at most qc_h points of Z, leaving at least c_h candidates. A valid exchange therefore exists in an unsaturated actual group. Its insertion has zero marginal ϕ, while deletion of x cannot increase ϕ.

These exchanges move the entire A-part into U. For $h = 1$ this finishes the proof. For higher rank, apply induction successively to the children. If W is the final returned core, a child operation cannot create a point of $(B \cap A) \setminus W$ and removes every such point in that child. All initially bad points lie in the union of the child ground sets, so none remain after all children are processed. This also justifies overlapping children and proves the invariant.

Size and running time. The output size is bounded by L_d, where

$$L_1 = T_1, \qquad L_h = T_h + \binom{T_h}{h-1} L_{h-1} \quad (2 \leq h \leq d).$$

All these quantities depend only on d, k. Recursion depth is at most d, branching is parameter bounded, and each node scans the input and constructs its children using parameter bounded sets in independence tests. Hence its oracle and bit cost is $f(d,k)\,\mathrm{poly}(|E|)$. Enumerate subsets of the output of size at most q, check all at most 2^q failures, minimize ϕ, and complete the quotas using original elements. The invariant preserves an optimum, and the enumeration has parameter bounded size. This proves Theorem 5.1.

Corollary 5.2. *For arbitrary directed state graphs, minimum unit-cost dedicated-actuator placement tolerating at most k actuator failures is fixed-parameter tractable in the maximum-matching deficiency d plus k, with at most one actuator per state.*

Proof. Use the dual of the state matching transversal matroid, whose rank is d. Surviving matching feasibility is spanning in this dual, and surviving accessibility is equivalent to selecting at least $k + 1$ states in each source SCC. The latter sets are disjoint, so Theorem 5.1 applies. □

The returned core need not contain a complete feasible actuator placement: quota completion may require many original elements. Thus the result is an FPT algorithm, not a claim of a bounded full-solution kernel. It does not resolve the best polynomial-time approximation ratio, and this quota extension makes no weighted or safe-element claim.

6 Relationship to earlier work and limits

The robust purchased-set objective differs from choosing an ordinary basis after failures. Bentert et al. [5] prove unweighted FPT in rank plus failure budget and explicitly pose the weighted question. Their proceedings Proposition 6 gives the layered cardinality bound, Lemma 8 the uniform-set exchange and basis-subset hyperplane count, and Lemma 9 the recursive core. These are inherited ingredients. Our weighted argument adds the ordered exchange invariant, recursion in both stopping cases, and separation of failure statuses. The sharper count is an application of Bollobás's classical set-pairs inequality [6], not a new set-pairs theorem. The same predecessor already gives polynomial weighted algorithms for rank at most two and for truncations of unit-capacity partition matroids.

The all-vulnerable case of Theorem 4.1 matches the weighted problem defined in Section 1.1 of [5]. Its Section 7 nonuniform question prints $\operatorname{rank}(M - F) = \operatorname{rank}(M)$; the selected-set formulation is $\operatorname{rank}_M(B \setminus F) = \operatorname{rank}(M)$. Lijun Cui reported in correspondence dated 5 November 2026 that Bentert confirmed the latter was intended and the former was a typographical error. We identify this as a secondhand report of a definition clarification, not as verification of our proof. The two printed predicates are not equivalent: in $U_{1,2}$, failure destroys the selected singleton $B = \{a\}$ but does not lower the rank of the full matroid. All our statements use the explicit selected-set definition (3). This selected-object interpretation also appears in flexible fault-tolerant network design [3].

Adjiashvili, Bindewald and Michaels [1], Theorem 7, give a 7/4-approximation for uniform cardinality node-robust assignment. Their repair perspective is a predecessor of the present approach. Their ICALP 2016 Theorem 5 [2] already gives 3/2 for uniform *edge*-robust assignment, buying assignment edges and tolerating an edge failure via ear decompositions. That is a different purchase/failure model. The proof in Section 3 instead uses fundamental cocircuit unions, maximum common-repair matching, and source quotas.

Bulk-robust optimization [4] charges purchased elements but supplies an explicit family of failure scenarios. Expanding all at most k failures can require $n^{\Theta(k)}$ scenarios, so polynomial time in that expanded list does not establish the present FPT bound. Weighted representative families [8] preserve independent extensions under their stated representation and matroid hypotheses; their predicate is not directly optimum preservation for a dependent purchased set under all failures. Likewise, the cheapest-surviving-basis preservers of Dey and Kavitha [7] optimize a different object, as the example above shows. These distinctions do not rule out alternative proofs using those methods.

The quota argument is separate from weighted spanning. An optimum robust core can be small although a feasible full placement is arbitrarily large. The completion objective and color caps preserve source choices without parameterizing their number. They make no weighted-quota or mixed safe/vulnerable quota claim. Conversely, the weighted theorem has no general source-quota guarantee. None of our results concerns repeated actuators at one state, arbitrary non-dedicated input columns, state or state-edge failures, or strong structural controllability.

The best polynomial-time approximation factor remains open. The three-state equality example limits the particular relaxation choice, not the underlying problem's approximability. Separate sparse-hardness research is not needed by, and is not reproduced in, the present proof package. Targeted source checks through 6 November 2026 identified the comparisons above but do not certify

priority or exhaust the literature.

Reproducibility. The accompanying repository provides dependency-free implementations and independent exhaustive, randomized, threshold, and nonrepresentable-matroid checks. The source records test commands and coverage. These finite audits validate implementations; they do not replace proofs. The code exposes rank oracles, which can be simulated from independence queries by greedy basis construction.

AI assistance. OpenAI tools assisted with developing and checking arguments, writing verification code, and preparing this manuscript. The proofs and code are provided for human review. Tool-assisted checking is not a substitute for independent mathematical peer review.

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